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A Neural Network Model to Estimate Aircraft Fuel Consumption

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ABSTRACT

The purpose of this paper is to present a simplified method to estimate aircraft fuel consumption using an artificial neural network. The models developed here are can be implemented in fast-time airspace and airfield simulation models. A representative neural network aided fuel consumption model was developed using data given in the aircraft performance manual. The data used in this study was applicable to the Fokker 100 aircraft powered by Rolls-Royce Tay 650 engines. A second data set was applied to the SAAB 2000 turboprop aircraft with good results. The methodology can be extended to any type of aircraft including piston and turboprop type vehicles with confidence.

The neural network was trained to estimate fuel consumption of an example aircraft. Results were compared to the actual performance provided in the aircraft performance manual and found to be accurate for possible implementation in fast-time simulation models. The result from the neural network model was compared with analytical models. The results of this study illustrate that a three-layer artificial neural network with nonlinear transfer func-tions can accurately represent complex aircraft fuel consumption functions for climb, cruise and descent phases of flight.

INTRODUCTION

Aircraft fuel consumption is a relevant issue in the planning and analysis of aviation operations in the National Airspace System (NAS). While fuel prices today represent only a fraction (i.e., 16-22%) of the Direct Operating Cost (DOC) of typical aircraft, they still constitute an important expenditure for airlines and general aviation operators. Based on this premise this paper attempts to develop a suitable method to estimate aircraft fuel consumption using an artificial neural network approach.

Fast-time simulation models have been used in the past decade as part of airport and airspace capacity studies. For example, in the late seventies the Federal Aviation Administration (FAA) developed SIMMOD - the airspace and airfield simulation model to address an internal need to estimate fuel consumption in airspace and airfield operations. This model has been widely used by internal groups within the FAA and over fifty users worldwide have some practical experience using this model.. More recently developed models such as the Total Airspace and Airport Model (TAAM) and the

Reorganized Analytical Modeling Systems (RAMS) use modern data structures to predict individual aircraft operations at airports and in airspace networks including some form of fuel consumption estimation. SIMMOD for example, utilizes a fuel consumption post processor that computes the fuel consumption of an aircraft given a flight profile using multivariate regression techniques (2,3). Others like TAAM employ complex table look-up functions to interpolate fuel consumption in the simulation. RAMS employs an explicit aircraft performance model derived from actual aircraft fuel consumption data.

Over the years, fast-time airspace and airport simulation models have been both acclaimed and criticized. In the past seven years the FAA has performed sixteen airport/airspace capacity enhancement studies where SIMMOD played a major role (i.e., was the modeling tool of choice) in the analysis (1). In this study all SIMMOD projects combined yield an expected delay savings of 4,703 million dollars in the horizon year (2003-2005). If one factors the life cycle cost savings of all major projects one can easily arrive to figures an order of magnitude higher. Other fast-time simulation models such as TAAM and RAMS have been used in more than forty airport and airspace studies recently. These studies while proprietary conclude some benefit at using these models.

It is important to note that in many instances these models have been used in numerous airport and airspace studies sponsored by airport authorities as well. In retrospect, fast-time simulation models have served, and continue to serve an important role in airport/airspace planning studies. This study provides and enhancement to fast-time simulation models by providing a computationally efficient and accurate method to predict aircraft fuel consumption. This improvement could be more relevant in the future when natural resource consumption become a more prominent variable in airport and airspace planning activities. The objectives of this paper are:

1. To investigate possible alternatives to fuel consumption modeling in fast-time simulation programs,
2. To use the information provided in the aircraft performance flight manual, along with neural network methodology to develop an accurate fuel consumption model, and
3. To determine whether neural network should be considered as a viable alternative in fuel consumption estimating applications.

METHODOLOGY

The neural network approach to aviation fuel consumption application is quite simple in principle. The aircraft fuel consumption data from the flight performance manual of individual aircraft are presented to the network. The network, by an iterative process, self-organizes and generates its own performance data (i.e., strength of connections between various elements of the network denoted as weights and biases are determined). This process is referred to as “network learning”. When sufficient amount of data are presented to the network, the network becomes a ”trained network” capable of estimating the performance of aircraft in fuel consumption. The next step is then to present random data to the network for generalization purposes and collect statistics on the errors observed between actual data and the outputs of model developed.

Artificial neural networks (or neural networks for short) are computational models broadly inspired by the organization of the human brain (4,5,6). The most important features of a neural network are its abilities to learn and to associate complex input and output mappings. This is done by presenting the system with a representative set of ex anamples describing the problem, namely pairs of input and output samples; the neural network will the n establish a mapping between input and output data. After training, the neural network can be used to recognize data that is similar to any of the examples shown during the training phase. The neural network can recognize incomplete or noisy data, an important feature that is often used for prediction, diagnosis or control purposes (7,8).

Fuel Consumption Data Sources

Data sets taken from the Fokker F-100 - a short range turbofan-powered aircraft were used to extract fuel consumption data to be used in the development of the neural network models (9). General specifications of this particular aircraft are shown in Figure 2. A scanner connected to a personal computer was used to digitize all flight performance charts contained in the flight manual. The flight manual consists of different charts relating fuel consumption to flight performance parameters such as Mach number, altitude, weight, and atmospheric temperature conditions (i.e., ISA, ISA + 10, etc.). Using a

data retrieval program all digitized maps were converted into usable table data depicting aircraft performance for various flight regimes. The typical segmentation of flight phases stated in the flight manual was maintained to preserve the accuracy of the tables as much as possible. Aircraft performance charts included takeoff and climbout, climb, cruise, descent and taxi profiles. An example of the original data is shown in Figure 3.

Design of the Neural Network Topology

Design of the appropriate neural network topology involves several important steps:

1. Choosing the appropriate neurons' transfer functions,
2. Basic decision about the amount of neurons to be used in each layer,
3. Selecting the amount of hidden layers.

Function approximation has been traditionally one of the most researched uses of neural networks. Typically, a two or three layer networks are sufficient to approximate complex functions with a finite number of discontinuities. In order to gain an insight as to how topology effects the outputs, tangent-sigmoid, logarithmic-sigmoid and pure linear neuron transfer functions were selected and tested for further investigation. A network topology study was conducted in order to find the most appropriate architecture neural network to fit aircraft fuel consumption parameters. Note that there are several hundred combinations of neurons and layers but for practical purposes we tested eight candidate topologies shown in Table 2. The training input data sets for this part of the experiment are 805 data points selected from the flight manual cruise section and then tested (or generalized) with 805 random data points selected from the same flight performance manual (9).

The network topology study is based on the number of floating point operations (flops) and output errors obtained for each network topology. Table 1 lists all the candidates selected and their corresponding computational performance. Based on the results shown in Table 1, the two best candidates are two layers with six neurons and three layers with eight neurons. The corresponding mean errors are 0.606% and 0.604% (for 10,000 epochs). Unfortunately, further testing showed that the two layer architecture does not converge to a minimum error during training with input data from the climb section after 10000 epochs. Therefore, the final neural network topology selected to solve the fuel con-

sumption problem employs a three layer model with eight neurons in the first two layers and one neuron in the output (third) layer. The corresponding transfer functions are logsig-tansig-purelin, respectively. This network is shown schematically in Figure 1.

Training the Neural Network

In order to simplify the analysis for development and training of the neural network models the MATLAB Neural Network Toolbox was employed. MATLAB is a general mathematical package produced by the Mathworks Company (10,11). This tool is very efficient to handle matrices and was used throughout this research project to handle data manipulation tasks and neural network computations.

The reader must understand that once a neural network is trained and properly calibrated its implementation becomes independent of the mathematical package used. Consequently although our approach was substantially simplified with the use of MATLAB the actual implementation of the neural network applied to aircraft fuel consumption is relatively easy to implement in any higher level programming language (including simulation languages like SIMSCRIPT II.5 and MODSIM) that handle arrays.

Throughout this project several programs or templates were developed in MATLAB to perform the following neural net computations:

1. Network training/learning.
2. Testing and evaluation of a trained network.
3. Implementation to estimate aircraft fuel consumption in typical aircraft trajectories.

For any given aircraft, the data was split into learning and testing sets. Learning sets are used to train the neural network to recognize patterns. Testing sets are random values of the input parameters used to generalize the neural network and to validate the outputs of the trained network. Each template requires the following inputs:

1. Number of inputs.
2. Value for the learning coefficient.

3. Number of processing elements (neurons) in the hidden layer and output layers.
4. Maximum number of cycles (epochs) for each run.
5. Required accuracy in the training procedure (i.e., sum of the squared errors for each run).

In general, all programs developed as part of this effort can be viewed as computational templates that are fully reusable for any number of aircraft. In this project two dissimilar aircraft were modeled to validate that the topology of the neural network employed could in fact characterize various performance envelopes for turbofan and turboprop powered aircraft. The Saab 2000 was employed as testing aircraft for the later assessment.

Selection of Training Algorithms

Based on the analysis performed with several transfer functions in the neural network the Levenberg-Marquardt algorithm was found to be the most efficient and reliable means to be used for this study. The neural network employed in the fuel burn evaluation model is based on non-linear optimization techniques. The objective of the optimization is to train the network parameters weights (w) and biases (b) so they can be adjusted in an effort to optimize the performance of the network. Neural networks are taught to accommodate changes in the weights and biases to appropriately reconfigure the output. During each training operation the error between the output and target becomes smaller until a minimized error goal is achieved. These weighs and biases are somewhat equivalent to the regression constants found in many nonlinear multivariate estimation models and thus can be easily incorporated in any programming environment that supports array manipulation.

Training Results

Flight information characteristics for the Fokker 100 aircraft (see Figure 2) were extracted from the flight performance manual for climb, cruise and descent conditions. Other flight phases such as take-off and climbout to 457 m. (1,500 ft.), taxiing and loiter (i.e., holding) conditions are simpler to analyze because the relationships are, in general, linear and can be approximated with a simpler

mathematical models (i.e., a simple regression model). These conditions can also be analyzed with lower order neural networks (2 layers and just 4-5 neurons in each layer).

The sizes of the various training data sets used in the neural network learning process are shown in Table 2. Note that all data sets used varied in length according to the characteristic nonlinear behavior observed. For example, the cruise phase, while non-linear in nature is more predictable with fewer points that the descent phase because the aircraft velocity profile changes more drastically in a descent from 11,280 m. (37,000 ft.) that cruising at the same altitude. A summary flowchart to estimate fuel burn is shown in Figure 3.

Examination of the flight manual data revealed that four climb profiles are typically employed in the operation of this aircraft for regular climbs (i.e., mach numbers ranging from a low 0.65 to a high speed climb at 0.75 Mach). In general, variations of fuel consumption are also very sensitive to climb weight and atmospheric conditions prevalent along the climb path. These are also represented in the flight manual (see Figure 4) through numerous charts. For this particular aircraft a low climb weight of 26,363 kg. (58,000 lb.) is possible with minimal payload (i.e., short stage lengths) up to a maximum climb weight of 44,545 kg. (98,000 lb.). Temperature profiles contained in the flight manual accounted for ISA (International Standard Atmosphere) and ISA+10 conditions. Note that in general the more information about varying temperature conditions is presented in the manual the more accurate predictions are possible using the neural network. A minimum data set to predict climb fuel consumption incorporates these four variables: a) pressure altitude at the top of climb, b) temperature, c) mach number, and d) aircraft weight.

For the climb phase, fuel consumption and distance estimation are trained for target altitudes ranging from 457 m (1,500 ft.) to 12,195 m. (40,000 ft.) For the cruise phase, specific air range (the distance covered for every pound of fuel consumed) is trained for cruise Mach numbers ranging from 0.3 to 0.77 at various altitudes and temperature conditions. Note that training data should be selected carefully such that a wide range of velocities and altitudes are included. Selection of training data is a very important step; whether the neural network can be used to predict fuel consumption accurately

depends on how well the trained network can generalize the input data. The climb fuel burn results are presented in Figure 4.

Validation of the neural network data

The cruise phase of the F-100 jet descibed above was chosen for comparison purposes. The parameter chosen for comparison was the specific air range(SAR), which is defined as the distance flown per unit of fuel consumed and so it is a measure of engine efficiency. A fairly accurate estimation of SAR is necessary for any cost analysis that may be performed. For testing purposes points were randomly chosen on the aircraft cruise performance curves in order to eliminate any bias towards the network. A set of 1000 random points were used with mach numbers ranging from 0.3 to 0.77, weights ranging from 62000 to 94000 lb and heights ranging from 8000 to 37000 feet. The results were plotted against real data and are shown in Figure 5.

Model Results

The generalization of a neural network involves testing various data sets into the trained neural network to assure the reliability of the fuel consumption estimations. Without doubt, this is one of the most important pieces in any neural network modeling effort. A well trained and constructed network should be able to predict fuel consumption conditions dissimilar to those used in the training procedure. The test results were evaluated based on the mean error and the standard deviation of errors. Standard hypothesis testing was used to ascertain the statistical significance of the results.

In this project it was decided that the number of data points in the generalization procedure should be equal or more than the number of points used in the training procedure. In all cases t-tests performed on the data sets demonstrated that the mean errors came very close to be zero and thus the null hypothesis was an all cases accepted (see Table 3).

Figures 5 shows the cruise phase fuel consumption results. In this figure a plot of Specific Air Range (SAR) versus cruise Mach number is shown. This representation is typical in flight performance manuals of high performance aircraft. This figure encompasses many charts and diagrams contained in the

flight performance manual and is also known as the “flight envelope” of the aircraft. In the modeling procedure data points were selected so as to include all possible flight conditions (including high-speed cruise conditions). The errors between the estimated and actual fuel burn are shown in Figure 6. Note that cruise fuel burn predictions are fairly accurate with a mean estimation error of -0.034% and a standard deviation of 0.334%. This is a result of the near quadratic behavior of SAR with Mach number.

Implementation Issues in Fast-time Simulation Models

The implementation of the neural network model can be extended to existing fast-time simulation models such as SIMMOD, RAMS or TAAM. The basic requirements to implement a neural network model are much less restrictive than those found today in these programs that typically rely on table look-up functions or multivariate nonlinear regression techniques. Schilling (12) demonstrated that neural network approaches can be conducted within reasonable CPU consumption resources without slowing down the simulation. For example, in the present implementation of fuel burn algorithms in SIMMOD the spacing between data points in the flight trajectory has to be tightly adjusted to less than 2,000 ft. (in the vertical dimension) to maintain reasonable accuracy. The proposed neural network model can be implemented with larger vertical spacing requiring less computational effort while maintaining a good level of accuracy. This is because the fuel consumption model using the neural network was generalized with absolute fuel consumption statistics representing altitude changes from sea level up to the cruising altitude. In this fashion large altitude changes are captured more accurately.

The existing input/output file structure used in fast-time simulation models can be utilized in the implementation of the neural network algorithm. Trani and Wing-Ho (13) have proposed modifications to the input file structure of SIMMOD to read weight and biases of the neural network files. This operation is analogous to reading large numbers of aircraft-engine specific constants as currently done in SIMMOD or in the table look-up functions of other models. A new set of files can be created to store weights and biases for a large aircraft population (100+ aircraft) to compute fuel consumptions for every phase of flight.

CONCLUSIONS AND RECOMMENDATIONS

A representative neural network aided fuel consumption model was developed using data given in the aircraft performance manual. The neural network was trained to estimate fuel consumption of an example aircraft. Results were compared to the actual performance provided in the aircraft performance manual and found to be accurate for possible implementation in fast-time simulation programs.

The following conclusions are derived from our analysis:

1. The information provided in the aircraft performance manual is a reliable source to obtain fuel consumption data of any aircraft. Along with neural network technology, a neural network aided fuel consumpti on model has been developed.
2. Results obtained from the neural network aided fuel consumption model show that a neural network with proper training is an accurate and efficient mean to calculate fuel consumption of fixed wing aircraft. The added benefit of this approach is that only the flight performance manual of the aircraft is needed to characterize the complete fuel burn behavior of the vehicle throughout its flight envelope.
3. A neural network is found to be a viable alternative in fuel consumption estimating application. The computational results obtained in this paper indicate that the neural network approach can be implemented in fast-time simulation models such as SIMMOD, RAMS, TAAM and future products where flight trajectories are described in terms of waypoints. Moreover, neural networks can approximate with good accuracy the complete performance of the vehicle (including climb, cruise, maneuvering, and decent) and simplify the implementation of realistic aircraft models without compromising proprietary aircraft data that is seldom made public.
4. The neural network is more accurate than existing analytical models such as Eurocontrol's Base Data of Aircraft (BADA) model. BADA has been used in many air traffic management studies by NASA and the FAA.

Recommendations

The model developed in this research project purely addressed the fuel burn and performance computations typical of fast time simulation models. A future enhancement to the model presented here is the extension to estimate thrust associated with a fuel burn flight condition. In simple terms thrust and fuel burn are related by a characteristic parameter called Thrust Specific Fuel Consumption (TSFC). This parameter is usually a complex function of mach number, temperature, pressure altitude, among other factors. Preliminary results obtained in our research indicate that thrust and TSFC can also be easily characterized using neural networks (we used a Pratt and Whitney JTD9-7R engine for this purpose) and thus thrust values can be obtained from operational simulation models to support noise studies.

The evolution of future airport and airspace models is likely to implement fuel consumption models as an integral part of the analysis and not as a post-processor module as currently done in most fast-time simulation models. In an environment where scarce economic resources are important it is perhaps inadmissible to forget the costs associated with aircraft operations in the National Airspace System (NAS). Many of the multi-million dollar decision making processes taking place today use highly aggregated cost economics (i.e., assigning average hourly costs per dissimilar aircraft populations). These analyses provide a crude approximation that can improved with models such as those proposed in this paper.

Recommendations for future research are:

a) Test the implementation of neural networks to predict fuel consumption for general aviation aircraft. This should be done to ensure that out network topology is robust and applicable to piston engine aircraft.

b) Integrate the model developed within the current structure of existing fast-time simulation models. This step should require little work since we have tested the algorithms in C and MATLAB for full aircraft trajectories.

c) Validate the model for a large aircraft database.

This last recommendation is a critical step if fuel burn is ever to be used by airspace and airport planners in a reliable fashion. Ironically, some fast-time simulation models such as SIMMOD were originally developed as a fuel consumption prediction tool. Yet few people today employ this model for this purpose because the fuel consumption data base is very small compared to the number of aircraft modeled operationally (only 17 aircraft are actually represented in terms of fuel consumption parameters).

ACKNOWLEDGEMENTS

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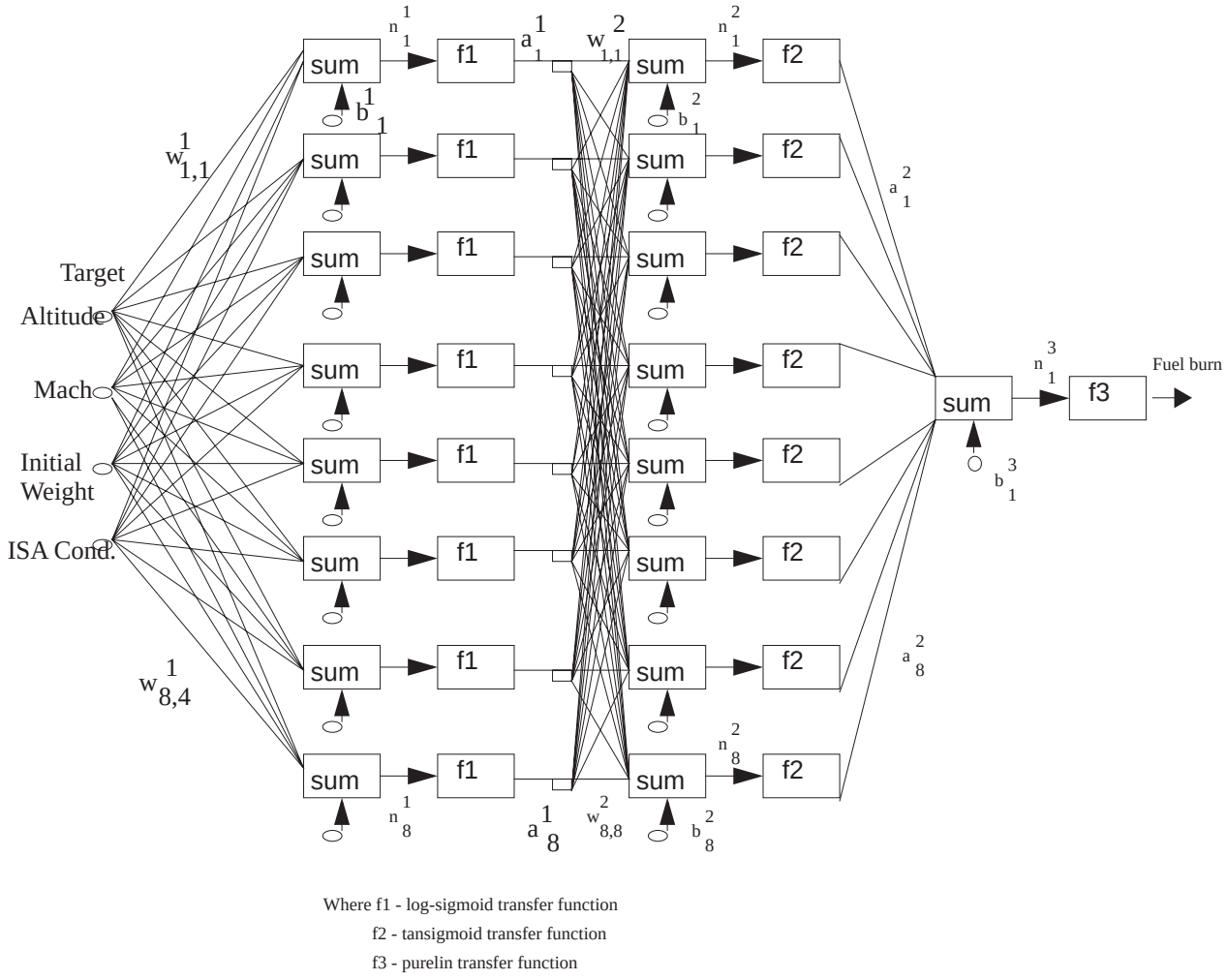
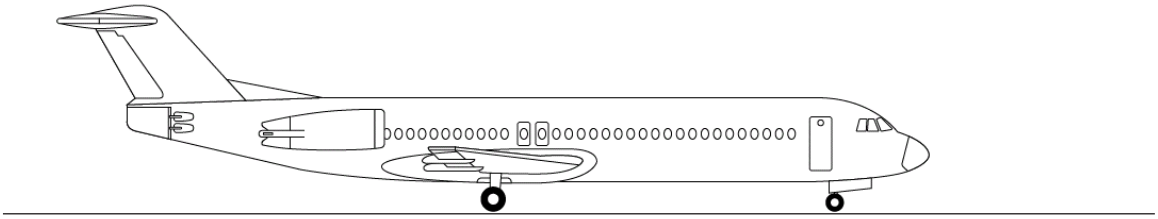
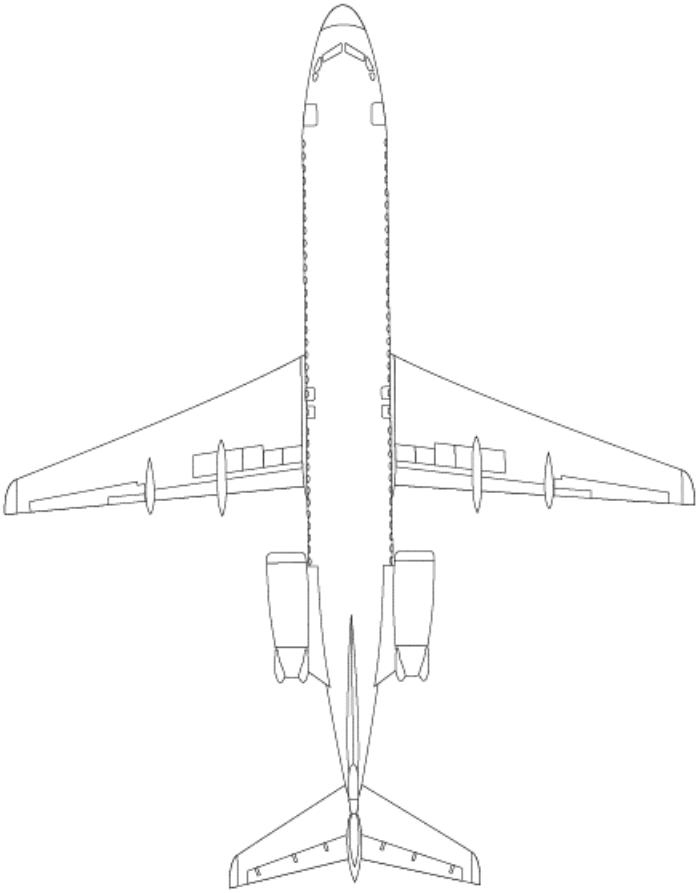


FIGURE 1. **General Three Layer Neural Network to Predict Fuel Consumption.**





Aircraft Characteristic	Value
Wingspan	28.08 m
Length	35.33 m
Overall height	8.5 m
Cruising speed	Mach 0.75
Approach speed	64.3 m/s
Service ceiling	12,000 m
Landing field length	1,520 m
Takeoff field length	1,830 m
Range	3,100 km
Powerplant	2 x RR Tay 650

FIGURE 2. Fokker F-100 Aircraft Layout and General Characteristics.

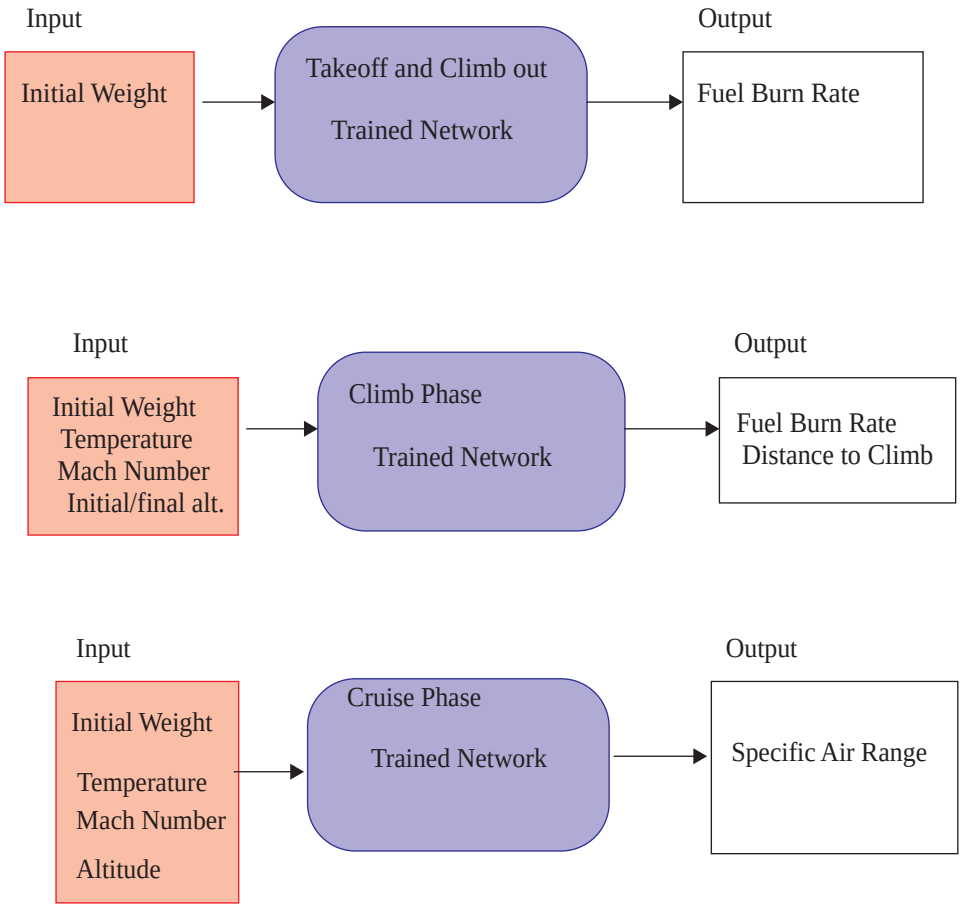


FIGURE 3. **Fuel Burn Estimation Procedures.**

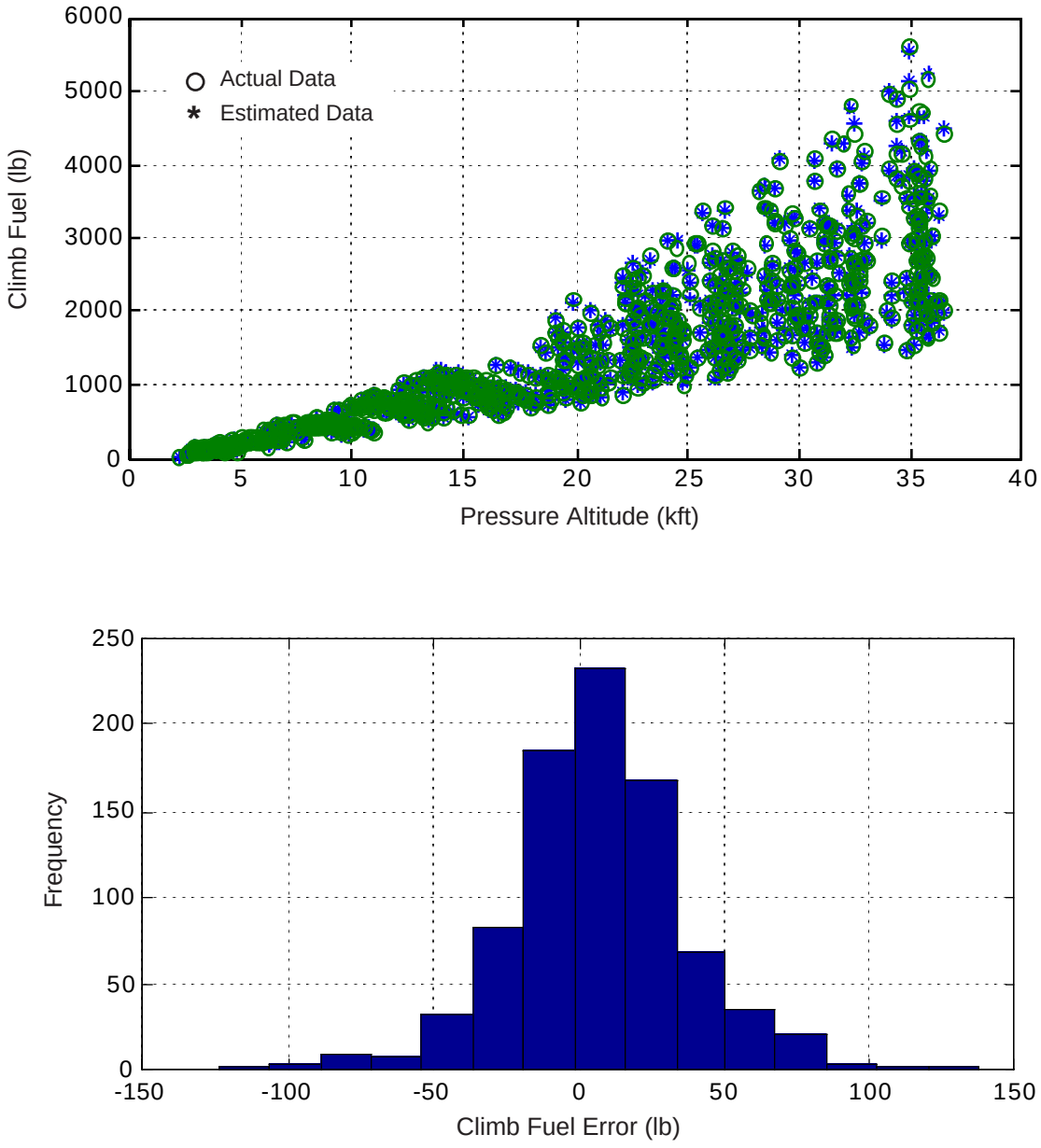


FIGURE 4. **Estimated and Actual Climb Fuel Results.**

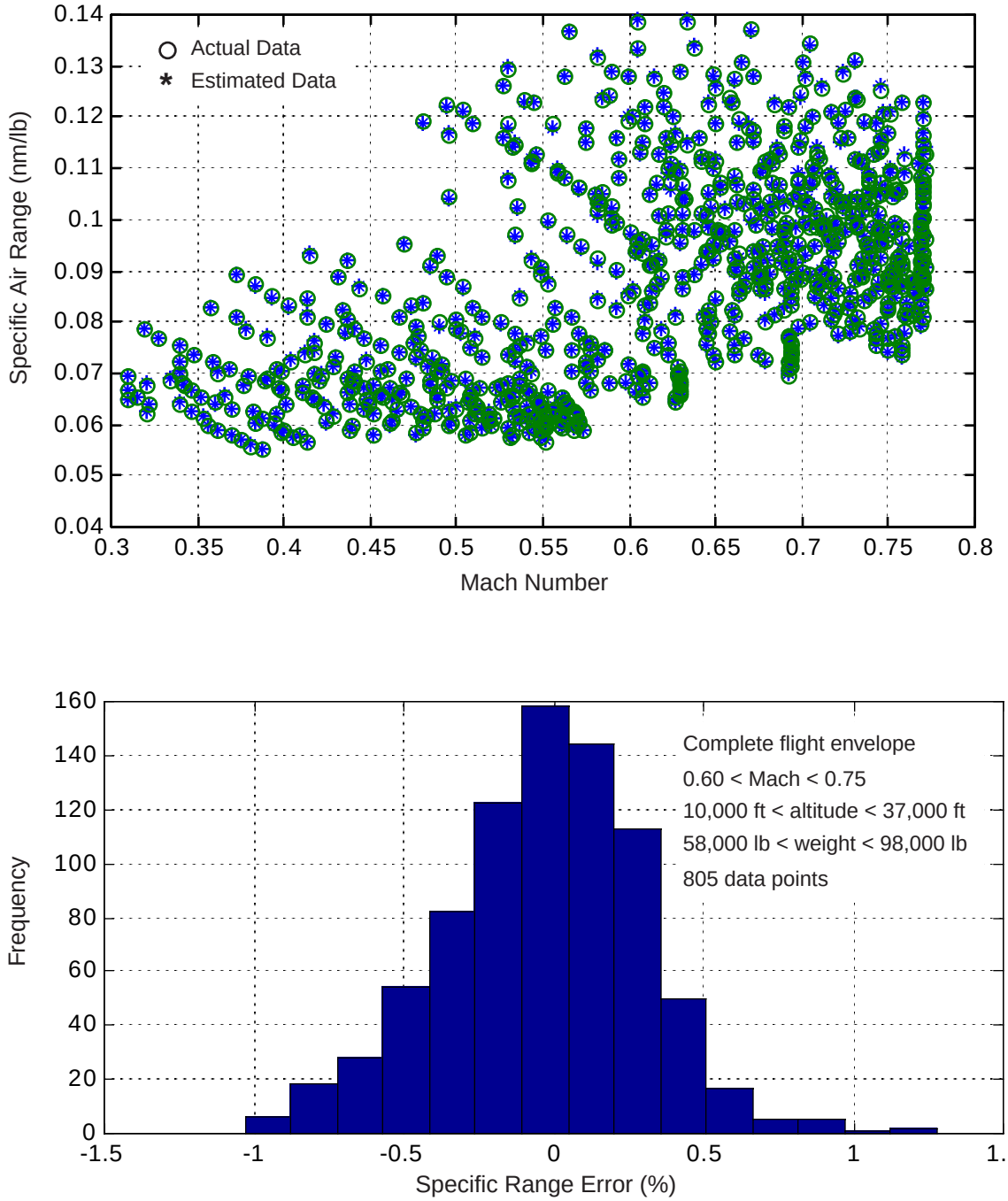


FIGURE 5. **Cruise Fuel Consumption Model Results.**

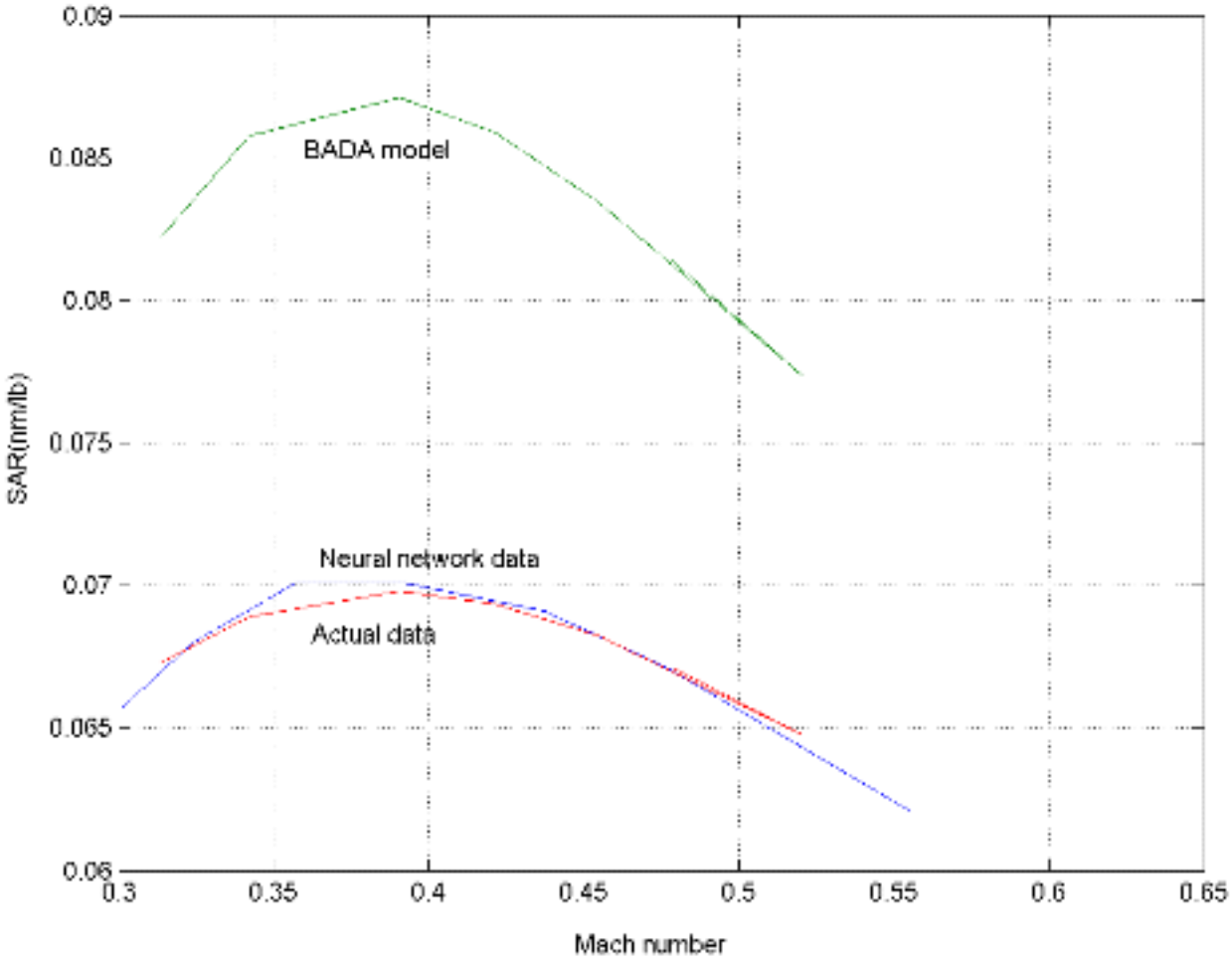


FIGURE 6. Comparison of the Neural Network Model with the BADA Model.

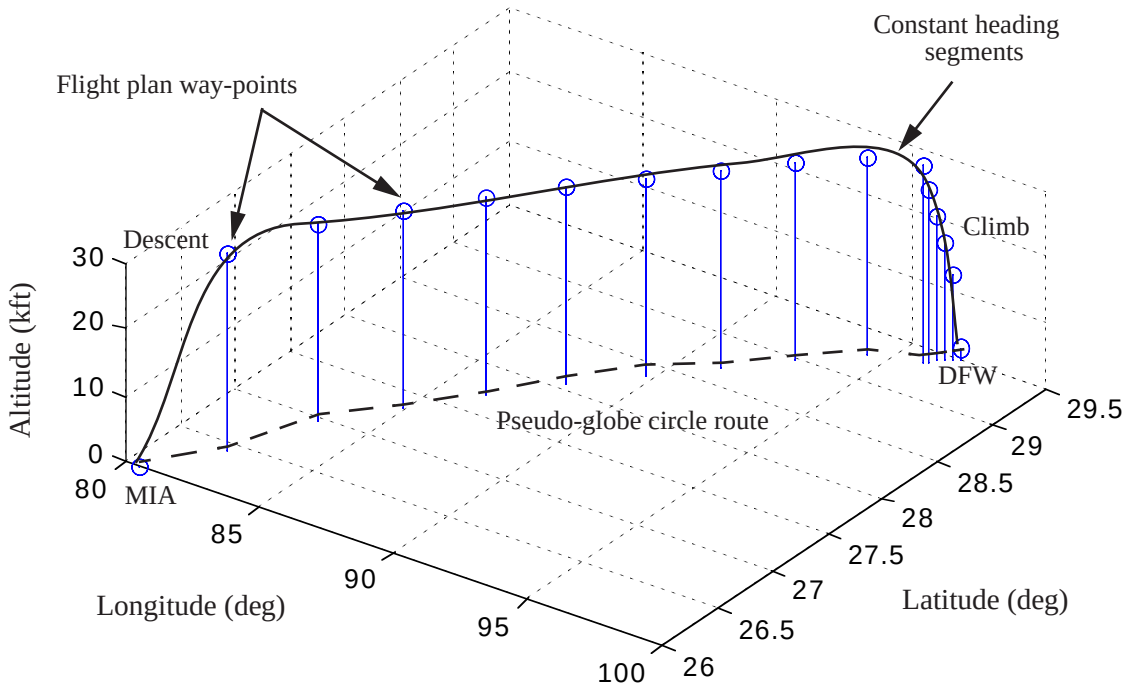


FIGURE 7. **Sample Flight Plan Profile.**

TABLE 1. Results of the Topology Sensitivity Study.

Number of layers	Number of neurons	Transfer functions	Mean Relative Error (%)	Standard Deviation of Error (%)	Floating point operations
2	4	tansig-purelin	0.611	0.0282	4.667E+08
2	6	tansig -purelin	0.606	0.0281	1.631E+09
2	8	tansig - purlin	0.628	0.0288	8.258E+08
3	4	logsig-tansig-pure-lin	0.617	0.0288	1.563E+10
3	6	logsig-tansig-pure-lin	0.610	0.0280	1.809E+10
3	8	logsig-tansig-pure-lin	0.604	0.0279	7.687E+08
3	10	logsig-tansig-pure-lin	0.656	0.0290	2.076E+09
3	12	logsig-tansig-pure-lin	0.667	0.0295	1.271E+09

TABLE 2. Summary of Neural Network Training and Testing.

Flight Phase	Number of Training Points	Number of testing points	Input Parameters	Output
Takeoff and Climb-Out	8	N/A	a) ISA Cond. b) Initial Weight (1000 lb.)	Fuel Burn Rate (lb/min)
Climb to Cruise Altitude	852 (Fuel) 854 (Distance) Total 1706	852 (Fuel) 854 (Distance) Total 1706	a) Initial Weight (1000 lb.) b) ISA Cond c) Mach Number d) Target Altitude (1000 ft)	a) Fuel Burn (lb.) b) Distance to Climb (nm)
Cruise	805	805	a) Cruise Mach Number b) Cruise Weight (1000 lb) c) Cruise Altitude (1000 ft)	Specific Air Range (nm/lb)
Descent	1210 (Fuel) 288 (Distance) Total 1498	1210 (Fuel) 288 (Distance) Total 498	a) Initial Weight (1000 lb) b) ISA Cond c) Mach Number d) Target Altitude (1000 ft)	a) Fuel Burn (lb) b) Descent Distance (nm)

TABLE 3. Statistical Results of Neural Network Model Calibration.

Flight Phase	Mean Error (%)	Standard Deviation (%)	Hypothesis Test (α = 0.01)
Climb			
• Distance	0.377	0.305	Accept
• Fuel	1.026	0.190	Accept
Cruise Specific Air Range	-0.034	0.334	Accept
Descent			
• Distance	1.760	1.860	Accept
• Fuel	1.423	1.177	Accept

TABLE 4. Flight Plans Used in the Correlation of the Neural Network Model (ISA+10).

Flight	Cruise Flight Level (FL)	Distance (nm) / Time (hr)	Flight Manual Fuel Burn (lb)	Neural Net Fuel Burn (lb)	Percent Difference (%)
ROA[a]-MDW[b]	280	448 / 1:08	6,457	6,546	1.37
	310	448 / 1:10	6,360	6,330	0.46
MIA[c]-DFW[d]	310	972 / 2:24	11,851	11,865	0.12
	350	972 / 2:13	11,510	11,544	0.29
ROA-LGA[e]	290	352 / 0:57	5,298	5,260	0.71
	330	352 / 0:58	5,343	5,429	1.61
ATL[f]-MIA	290	518 / 1:20	6,990	7,047	0.80
	330	518 / 1:21	7,009	7,082	1.04
ATL-DCA[g]	290	475 / 1:13	6,549	6,584	0.54
	330	475 / 1:14	6,590	6,654	0.97
ROA-ATL	280	310 / 0:51	4,938	4,998	1.30
	310	310 / 0:51	4,941	4,933	0.15

a. ROA - Roanoke Regional Airport (Virginia)
b. MDW - Midway Airport (Illinois)
c. MIA - Miami International (Florida)
d. DFW - Dallas-Forth Worth International (Texas)
e. LGA - Laguardia Airport (New York)
f. ATL - Atlanta Hartsfield International Airport (Georgia)
g. DCA - Reagan National Airport (Virginia)